THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 13: **Earth as a System: Energy, Matter and Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A student claims 'CO2 is harmless as it is natural'. A good reply is: 1

(a) it is never found in air
(b) it is always bad
(c) it is poison to plants
(d) its increase beyond natural levels enhances the greenhouse effect

2. Satellites orbiting in the thermosphere are used for: 1

(a) growing crops **(b)** weather and communication
(c) swimming **(d)** digging

3. Assertion (A): Deforestation contributes to the increase in atmospheric CO2. Reason (R): Trees remove CO2 through photosynthesis and store carbon. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. A car running 12000 km per year at 150 g CO2/km emits: 1

(a) 18 tonnes **(b)** 0.18 tonnes **(c)** 180 kg **(d)** 1.8 tonnes

5. Which action reduces both CO2 emissions and water pollution? 1

(a) dumping waste **(b)** using diesel
(c) reducing the use of chemical fertiliser **(d)** burning crop residue

6. The speed of sunlight in vacuum is 3 x 10^8 m/s. A signal from Mars (distance 2.25 x 10^11 m) takes: 1

(a) 75 s **(b)** 750 s **(c)** 7.5 s **(d)** 7500 s

7. Assertion (A): The ozone layer is useful to humans. Reason (R): It absorbs the harmful UV rays of the Sun. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. A 100 W bulb used for 5 hours a day for 30 days uses: 1

(a) 15 kWh **(b)** 0.15 kWh **(c)** 150 kWh **(d)** 1.5 kWh

9. **Assertion (A):** The ozone layer protects living organisms from harmful ultraviolet radiation. 1
Reason (R): The ozone layer lies in the troposphere, close to the ground.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Plants release carbon dioxide to the atmosphere during photosynthesis. 1
Reason (R): Photosynthesis uses carbon dioxide and water to make glucose and oxygen.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A greenhouse gas has a lifetime of 100 years and another of 10 years. Which will remain longer and what are the implications for the climate action? 2

12. State two reasons for the low albedo of the oceans. 2

13. Why is the CO2 level measured at Mauna Loa showing seasonal ups and downs with a rising trend? 2

14. Why do we use the term 'carbon neutral'? Give an example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how the ocean absorbs CO2 and the problem of ocean acidification. 3

16. The CO2 level is increasing at 2.5 ppm per year. From 420 ppm, find the level after 20 years and the percentage increase. 3

17. What is a carbon sink? Name three natural carbon sinks and one artificial. 3

OR

Describe how the nitrogen cycle can be disturbed by the use of fertilisers and how organic farming reduces the effect. 3

18. Explain how the monsoon shows the link between the ocean, the atmosphere and the land, and how a warmer ocean might change it. 3

OR

Explain the term 'ecological footprint' and how one can compare two lifestyles. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A rooftop of area 100 m^2 in Chennai receives 0.8 m of rain in a year. The residents plan rainwater harvesting and also read that burning of fossil fuels is increasing the concentration of carbon dioxide in the atmosphere, leading to global warming. 5
(i) Find the volume of rainwater falling on the roof in a year.
(ii) Express this volume in litres ($1\ m^3 = 1000$ L).
(iii) Name one greenhouse gas other than carbon dioxide.
(iv) Explain how an increase in greenhouse gases leads to a rise in the temperature of the Earth.

20. Plan an investigation on the effect of albedo on temperature using white and black cards or cans under a lamp or the Sun: the apparatus, variables, procedure, expected results and the link to climate. 5

OR

Discuss how the four spheres changed during the last glacial period and how the present warming differs, using the evidence from ice cores of CO2 concentration (180 ppm in glacial, 280 pre-industrial, 420 today). 5

END OF QUESTION PAPER — Total: 40 Marks